



Can't stop the Music

Napster may be history, but peer-to-peer file sharing is far from dead. A new generation of file sharing applications are keeping the flame alive, despite RIAA's attempts to stop the digital music blitzkrieg

The golden era for Napster buffs is over. The music stealing software of choice now uses a finger printing technology that prevents the trading of copyrighted material. The 50-million-strong service has been re-formed to a membership-based service—users will now pay for downloads. A dozen other services have gone under too, as legal battles mounted by the record industry take their toll.

But RIAA's (Recording Industry Association of America) attempts at filtering music are about as effective as trying to soothe a rioting mob with a copy protected Kenny G CD. Ironically, for every target the RIAA has fought, there seem to be three more that spring out of the shadows. The services that gave momentum to peer-to-peer (P2P) file sharing may be extinct, but P2P is thriving and is one of the hottest technology buzzwords today. The number of file-swapping sites has grown more than five-fold in the past year, despite legal efforts by music companies and software firms to shut them down. Nearly one in three of the most popular applications in **Download.com** are for P2P services.

Power to the people

One of the most popular second-generation P2P applications is Morpheus, distributed by StreamCast Networks. Since

its inception, 60 million have already downloaded Morpheus and a large number of them are heavy users. In the two weeks after StreamCast upgraded its version of Morpheus onto the open source Gnutella technology, 25 million users downloaded the new version. This proved to be of help to the Gnutella network as well—the number of people sharing files on the Gnutella network jumped from 100,000 to 500,000. Limewire, Morpheus, Bearshare, and many other P2P applications now share the same backbone, Gnutella. Users can share, search, and download not only all types of music, but also movies, pictures, games and text documents.

Justin Frankel and Tom Pepper at Nullsoft developed the original open source Gnutella client—the same company that created Winamp, the world's most popular MP3 playback software and Shoutcast, the number one Internet radio application. Many different companies, individuals and groups have gone on to modify and evolve the Gnutella protocol.

Gnutella is the record industry's worst nightmare, built to survive both court rulings and nuclear war. While P2P services based on centralised servers may be vulnerable to lawyers and court rulings, decentralised servers such as those based on the Gnutella technology can-